

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay.
- Best TSPLIB holdout gap: tsplib_failure_replay.
- Best synthetic holdout gap: tsplib_failure_replay.

Run Metadata

- run_name: run_20260513_153347_r
- started_at_local: 2026-05-13 15:33:47
- finished_at_local: 2026-05-13 15:43:02
- duration_hhmm: 00:09
- duration_seconds: 555.411
- seed_offset: 17000
- replicate_label: r
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.914661	0.56	-0.027229
tsplib_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.879896	0.56	-0.028304
tsplib_failure_replay	failure	score_only	False	0.119843	0.0	0.076264	0.804813	0.76	0.018215
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.914661, and final complexity 0.56.
- Adaptation efficiency -0.027229 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.879896, and final complexity 0.56.
- Adaptation efficiency -0.028304 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.119843, synthetic 0.0, combined 0.076264.
- Accepted-epoch count 3, mean accepted code novelty 0.804813, and final complexity 0.76.
- Adaptation efficiency 0.018215 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.119843 across 7 instances; family means: ch=0.121576, kroD=0.081948, pcb=0.220292, pr=0.048345, rd=0.199241, st=0.045926.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3408, "family": "a", "name": "a280", "optimality_gap": 0.321442}, {"best_known_cost": 7542, "cost": 9753, "family": "berlin", "name": "berlin52", "optimality_gap": 0.293158}, {"best_known_cost": 629, "cost": 761, "family": "eil", "name": "eil101", "optimality_gap": 0.209857}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

Judge Appendix

Summary of Transfer Performance and Optimality Gaps by Condition

Condition	Replay Mode	Selection Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Complexity	Notes
tsplib_failure_replay	failure	score_only	**0.119843**	**0.0**	**0.076264**	0.804813	0.76	Best heldout and synthetic performance. Clear transfer improvement over no replay and random replay by ~0.09 TSPLIB gap. Some decrease in code novelty compared to no replay/random replay.
tsplib_no_replay	none	score_only	0.213387	0.064916	0.159398	0.914661	0.56	Baseline with highest gaps across transfer and heldout sets. Highest code novelty and lowest complexity.
tsplib_random_replay	random	score_only	0.213387	0.064916	0.159398	0.879896	0.56	Same gaps as no replay, but slightly reduced code novelty. Random replay does not improve transfer metrics.
tsplib_failure_replay_compression	failure + compression	novelty_gate	0.213387	0.064916	0.159398	0.0	0.56	Transfer and heldout gaps identical to no replay. Code novelty collapsed to zero, possibly due to compression pressure.

Interpretation

- **Transfer & Optimality Gap:**

The **tsplib_failure_replay** condition shows clear and consistent improvement in both heldout TSPLIB and synthetic datasets, reducing mean transfer gaps roughly by half relative to no replay or random replay. According to the rules, these metrics are the main evidence for transfer and indicate the failure replay strategy benefits generalization and solution quality on unseen instances.

- **Code Novelty vs Transfer:**

Despite the improvement in transfer and optimality gaps, the **failure replay** condition exhibits a reduction in code novelty (mean ~0.80) compared to no replay (~0.91) and random replay (~0.88). This confirms that improved transfer does come with lower code novelty, as per instructions to report explicitly. However, failure replay with compression

pressure leads to zero code novelty and no transfer gains, indicating loss of effective adaptation.

- **Replay Modes:**
- **No replay** and **random replay** conditions show indistinguishable transfer performance and optimality gap, suggesting random replay alone does not improve transfer.
- **Failure replay** condition uniquely improves transfer metrics, confirming the benefit of targeted replay of failure cases.
- **Failure replay with compression** restricts novelty and degrades transfer, implying trade-offs between compression pressure and transfer gains.
- **Complexity:**

The failure replay condition produces a higher behavioral complexity (0.76) than other conditions (0.56), potentially indicating more elaborate solutions supporting better transfer and lower gaps.

Conservative Conclusion

- **Failure replay** is the only replay-aware method to demonstrate measurable improvements in holdout TSPLIB and synthetic TSP instance optimality gaps, confirming its positive effect on transfer.
- This improvement coincides with moderately reduced code novelty and increased behavioral complexity, consistent with a shift towards convergence on better quality solutions rather than exploratory novelty.
- Random replay and no replay perform equivalently, negating the efficacy of random sample reuse without failure focus.
- Compression-aware replay under failure replay conditions causes failure of transfer gains and collapses code novelty, suggesting detrimental effects from compression pressure in this setting.